

Positional Embedding

Advanced NLP M26

Manish Shrivastava + Aparajitha



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Transformer

Where are we?

Positional
Embedding

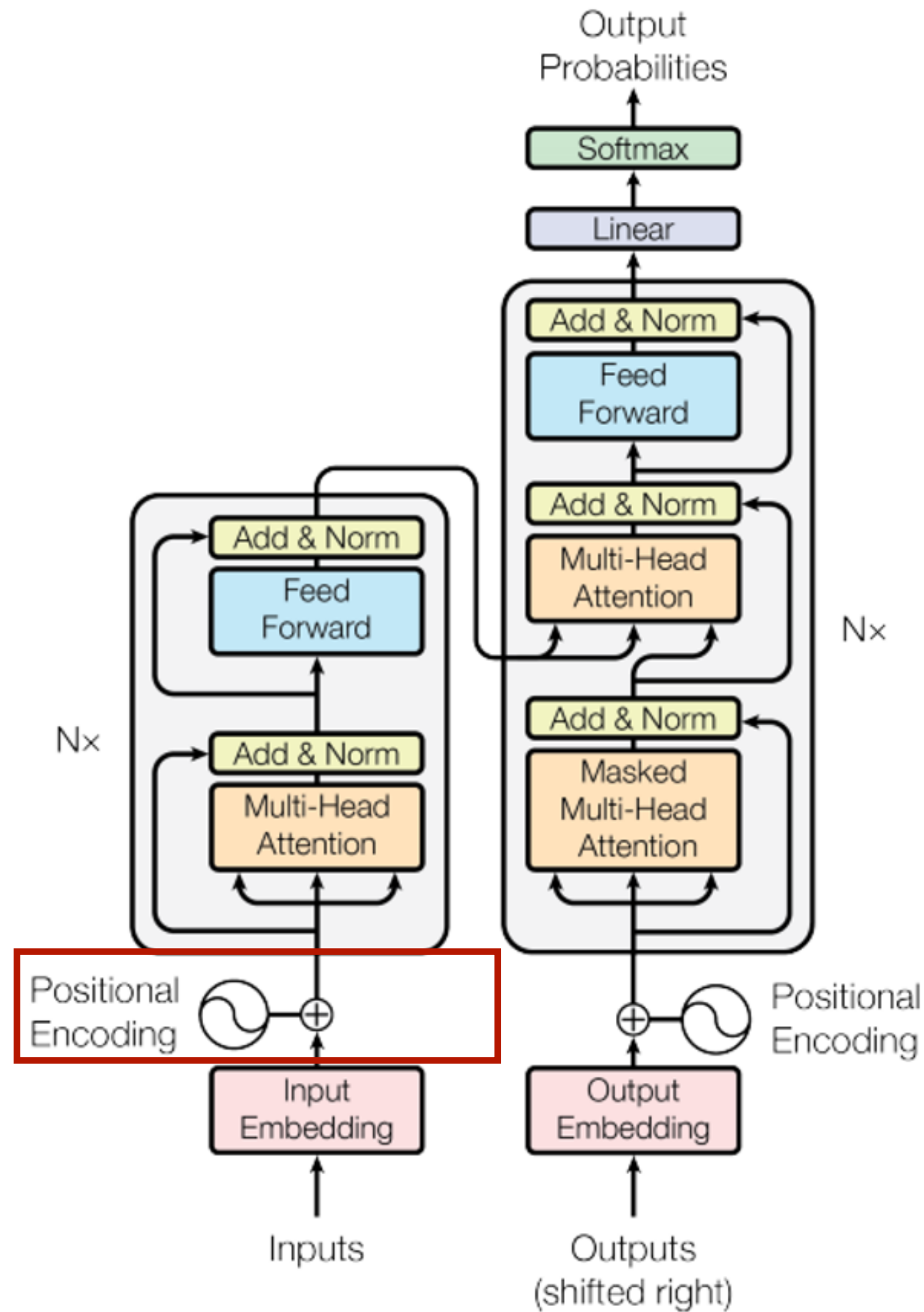


Figure 1: The Transformer - model architecture.

Word Position

Importance

- Languages - order of the sequence
 - No free order
 - Relative free order
- Noun/Verb Modifiers
- Causality
- Temporal



Absolute PE

Absolute PE

Approach

- Vectors with the same dimension as the word itself
- Each word gets a unique absolute embedding to show its position in the sentence



Absolute PE

Drawbacks

- Model just remembers the position vectors
- No information about relative positions
- Struggle to predict unseen (during training) positions



Sinusoidal PE

Sinusoidal PE

Approach

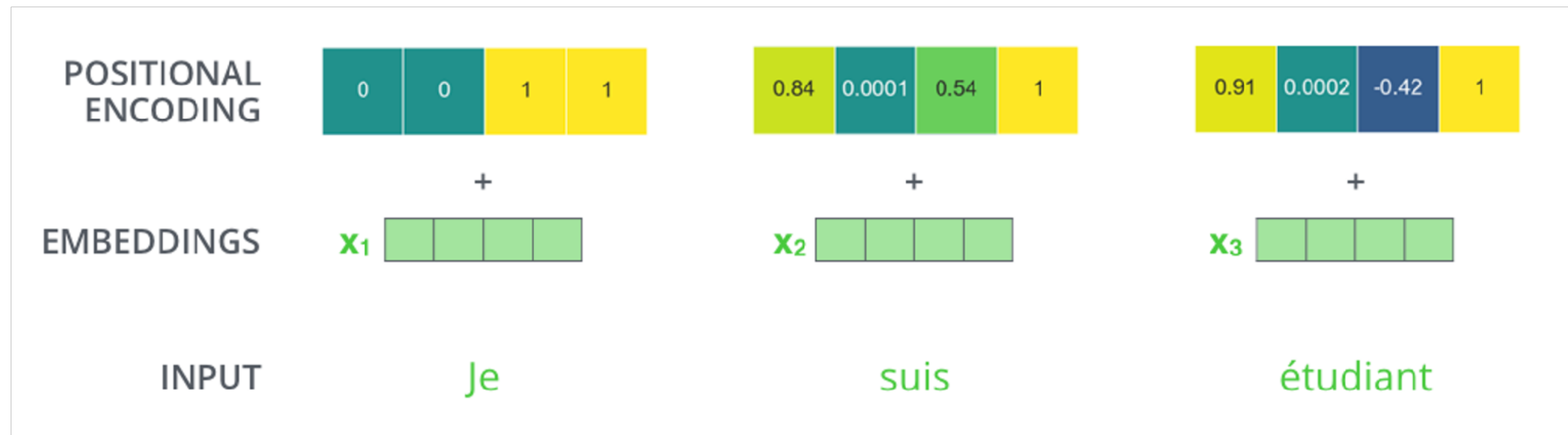
- Hypothesis: Allow the model to easily learn to attend by relative positions
 - For any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos}
- Added to the word embedding

Sinusoidal PE

Positional Encoding

$$PE_{(pos, 2i)} = \sin(pos / 10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i/d_{\text{model}}})$$



Sinusoidal PE

Gains

- Word position
- Similar performance as learned position embeddings
- May allow the model to extrapolate to longer sequences than the ones encountered during training

Sinusoidal PE

Drawbacks

- Limited generalization to sequence lengths much longer than those seen during training
- Fixed/pre-defined position; no learnable parameters



Relative PE

Relative PE

Approach

- Instead of individual token embedding; pairwise embedding
- PE calculated between two vectors
- Added to self-attention mechanism instead of token vector



Relative PE

T5 - Approach

- Simplified form of PE - a scalar is added to the corresponding logit used for computing the attention weights
- Shared position embedding parameters across all layers
- Each attention head uses a different learned position embedding

Relative PE

T5 - Approach

- 32 embeddings with ranges that increase in size logarithmically up to an offset of 128
- Beyond - assign all relative positions to the same embedding
- Given layer is insensitive to relative position beyond 128 tokens,
- Subsequent layers can build a sensitivity to larger offsets by combining local information from previous layers



Relative PE Gains

- Enhanced representation
- Better generalization at varied sequence length



Relative PE

Drawbacks

- Implementation Complexity
- Delay in training and inference
- Cant cache - compute new key-value pairs for every new token



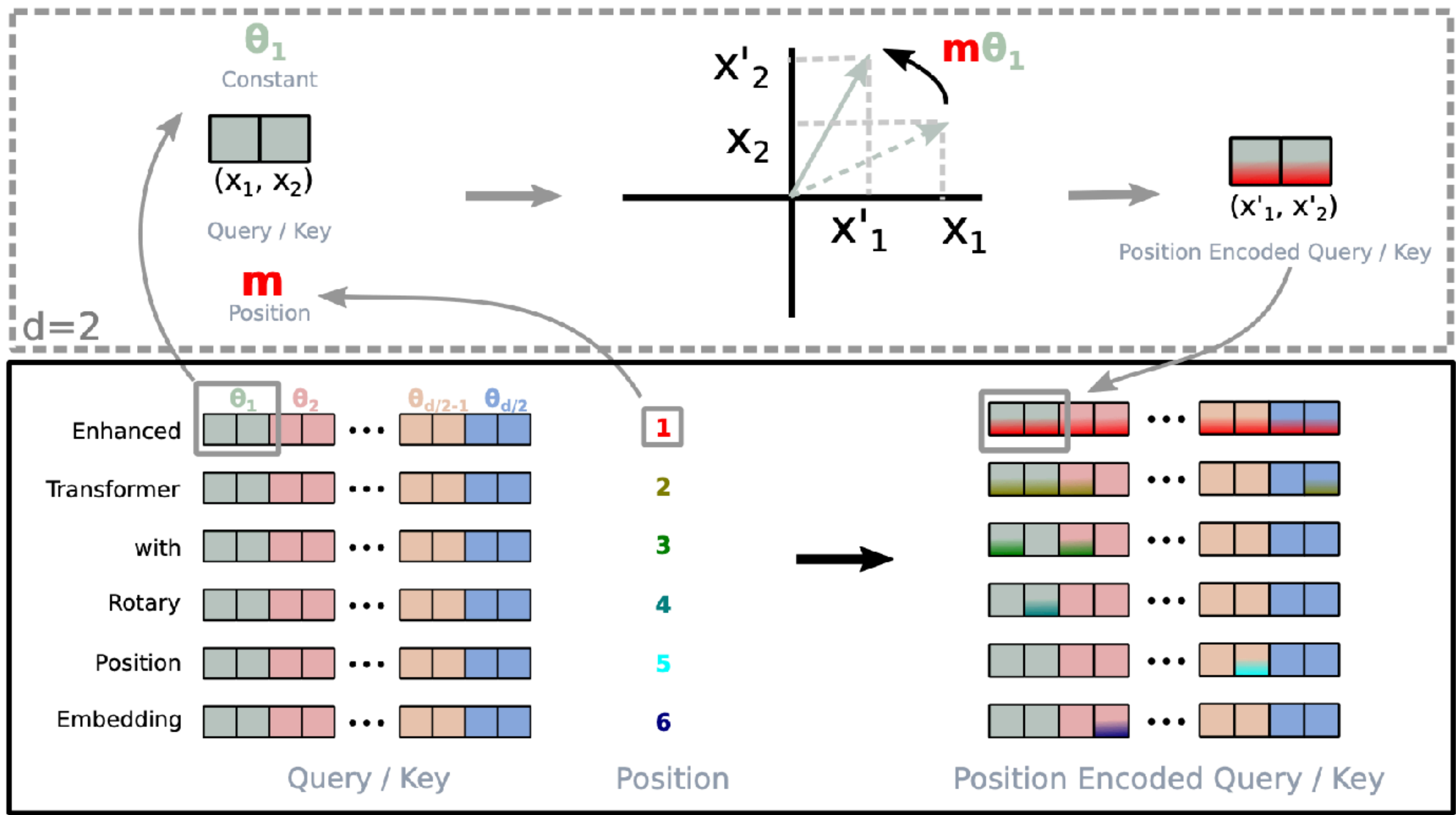
RoPE (Rotary PE)

RoPE

Approach

- Integrate positional information into the learning
- Encodes absolute position with a rotation matrix
- Incorporate the explicit relative position dependency in self-attention formulation
- Rotate the affine-transformed word embedding vector by amount of angle multiples of its position index

RoPE Implementation



RoPE

Approach - 2D Example

$$f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}$$

$$f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta}$$

$$g(\mathbf{x}_m, \mathbf{x}_n, m - n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}]$$

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$

RoPE

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \mathbf{R}_{\Theta, m}^d \mathbf{W}_{\{q,k\}} \mathbf{x}_m$$

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

rotary matrix with pre-defined parameters $\Theta = \{\theta_i = 10000^{-2(i-1)/d}, i \in [1, 2, \dots, d/2]\}$

RoPE

Gains

- Flexibility of sequence length
- Decaying inter-token dependency with increasing relative distances
- Capability of equipping the linear self-attention with relative position encoding
- Interpretability rotational approach offers intuitive geometric interpretation of how positional information influences the attention mechanism

RoPE

Drawbacks

- Implementation Complexity

NoPE (No PE)

NoPE

Approach

- No explicit positional embedding - not fixed, learned, or relative
- Causal mask gives autoregressive model an implicit sense of order
- Model learns to exploit the asymmetry to reconstruct positional information on its own via SGD

NoPE

Approach

- Causal Mask - a token at position t can only see tokens at positions $\leq t$,
- Implicit sense of direction
- Regular gradient-descent-based training learns to exploit it if it finds it beneficial for the optimization objective
 - When trained with SGD it mostly resembles T5's relative PE attention patterns

NoPE

Approach - Current Models

- SmolLM3 uses a mixed design and omits RoPE in every fourth layer.
- Kimi applies NoPE in its MLA global-attention layers
 - This lets MLA run as pure multi-query attention at inference and avoids RoPE retuning for long-context scaling
- Kimi K3 uses only NoPE, no RoPE layers at all



NoPE

Gains

- Simple - no additional computation
- Generalizes to longer sequences better than RoPE or absolute embeddings
- Theoretically can represent both absolute and relative positional encodings

